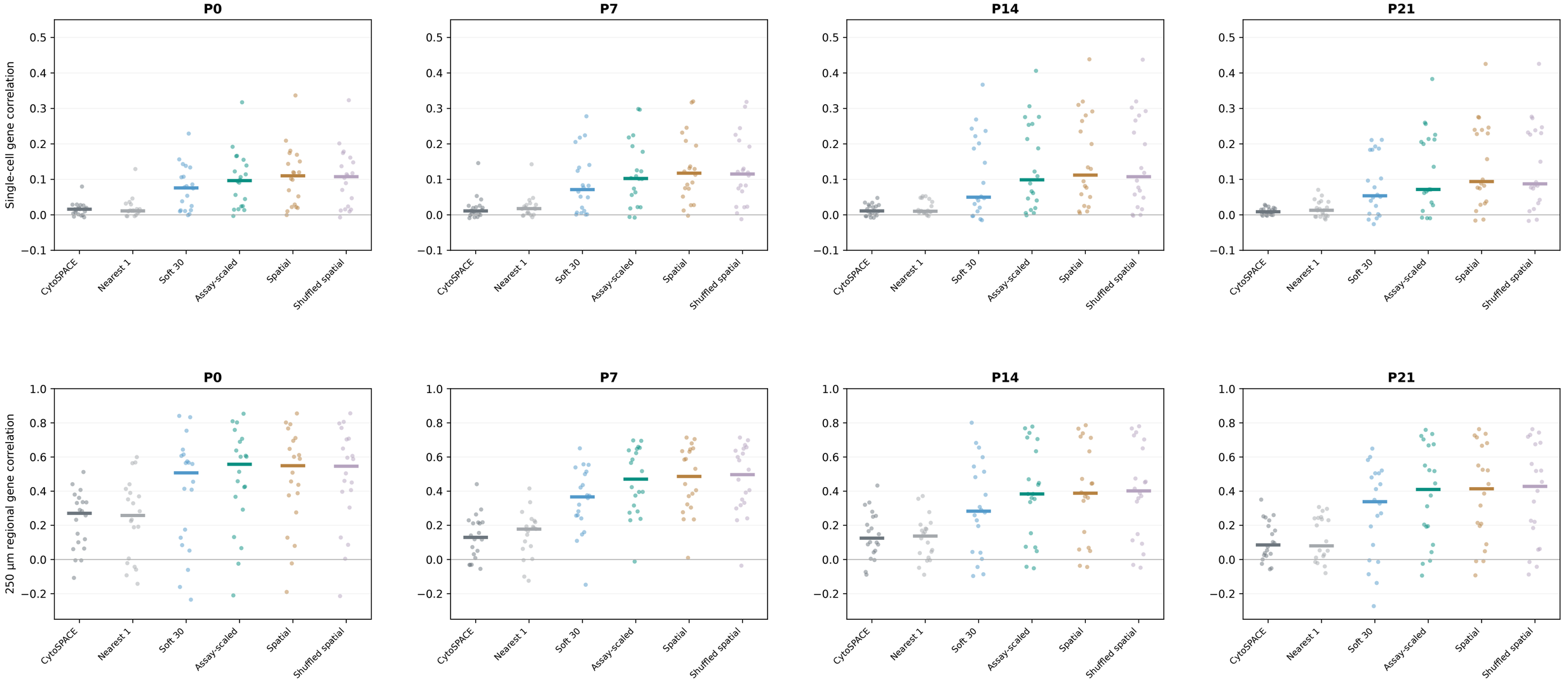
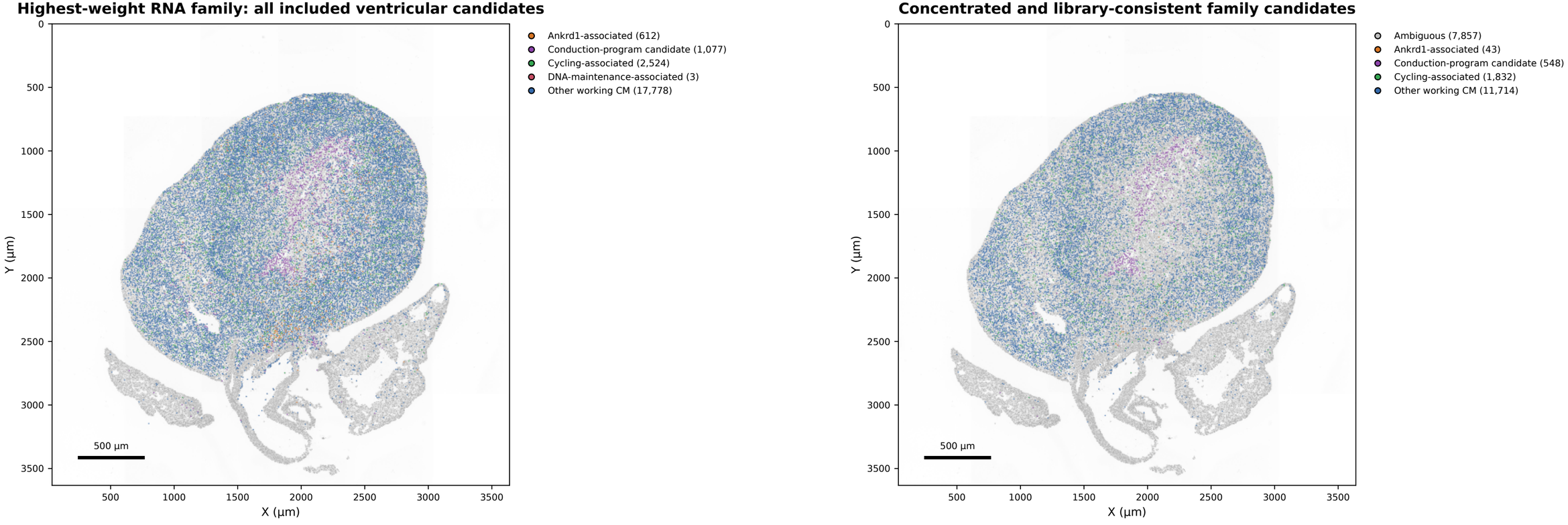
Prediction of 20 newly withheld genes: same cells and training genes for every method



Each point is one withheld gene; bars are medians. All 39 held-out genes were excluded from fitting and normalization denominators.
These are internal gene-recovery tests conditional on the frozen ventricular gate, not independent anatomical subtype accuracy. Shuffled spatial is a control.

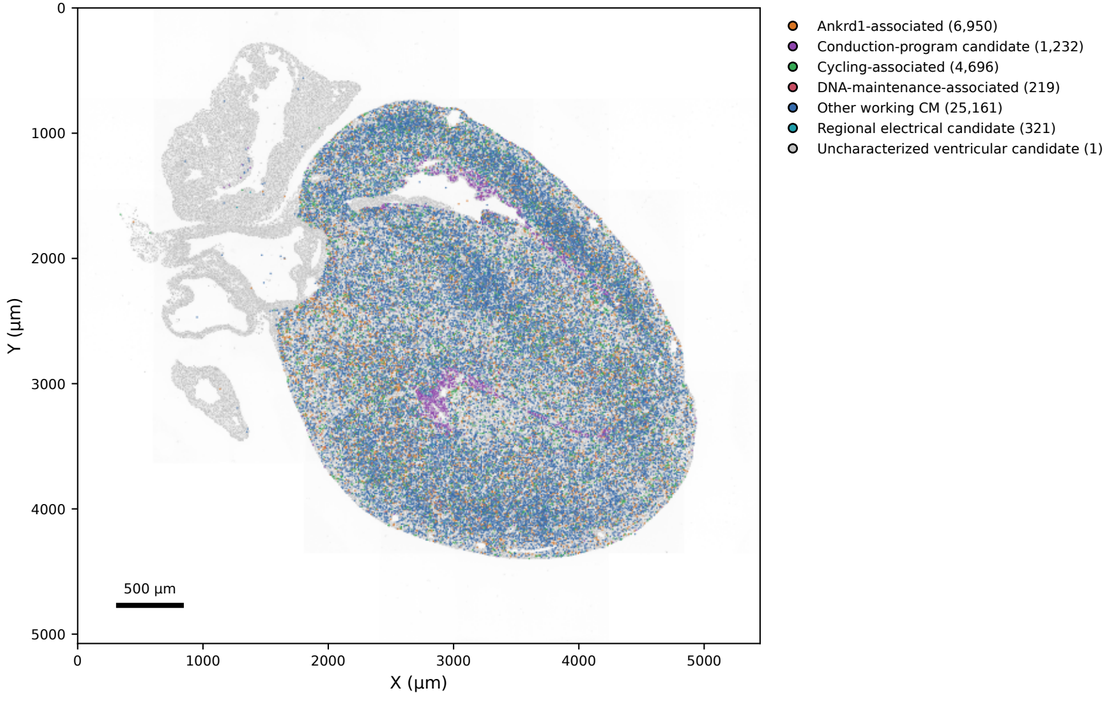
P0: unsmoothed weighted projection of ventricular RNA programs



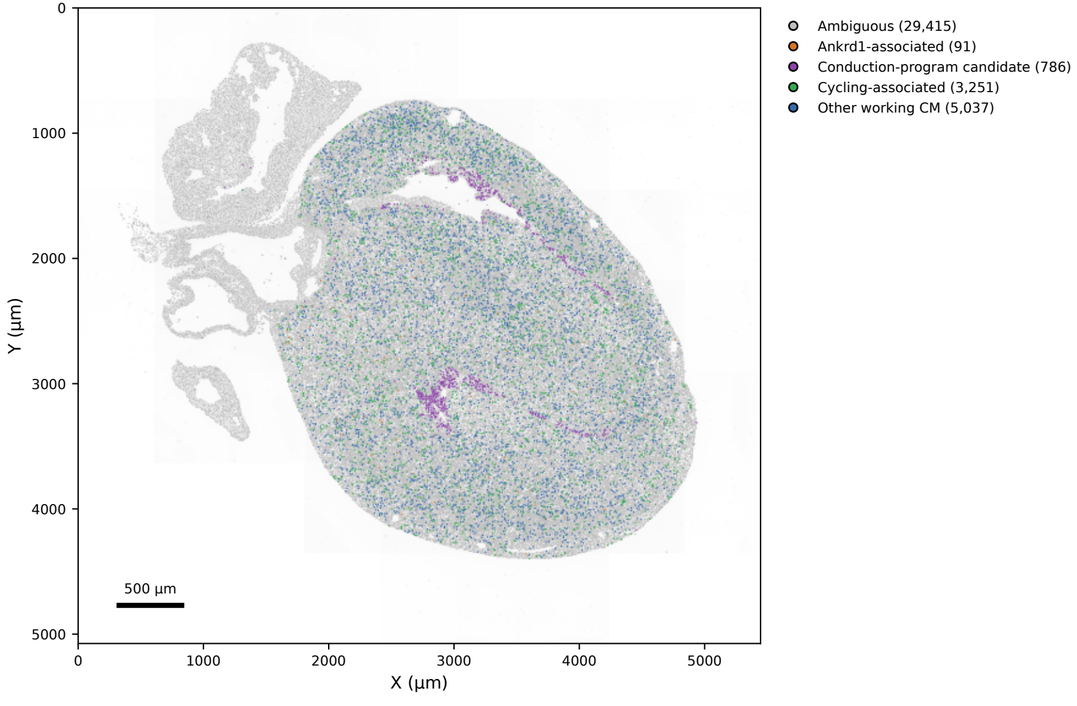
Original Xenium coordinates and counts are preserved. Left: strongest reference family; right: ≥0.70 neighbor weight and the same family from both RNA libraries.
Gray cells are ambiguous or outside the frozen cohort. Weight fractions are not calibrated probabilities. These flags differ from the previous CytoSPACE acceptance rules.
Conduction-program candidates are not verified Purkinje cells. Kcnd2/Irx5 are unassayed; neither is used to orient these sections.

P7: unsmoothed weighted projection of ventricular RNA programs

Highest-weight RNA family: all included ventricular candidates

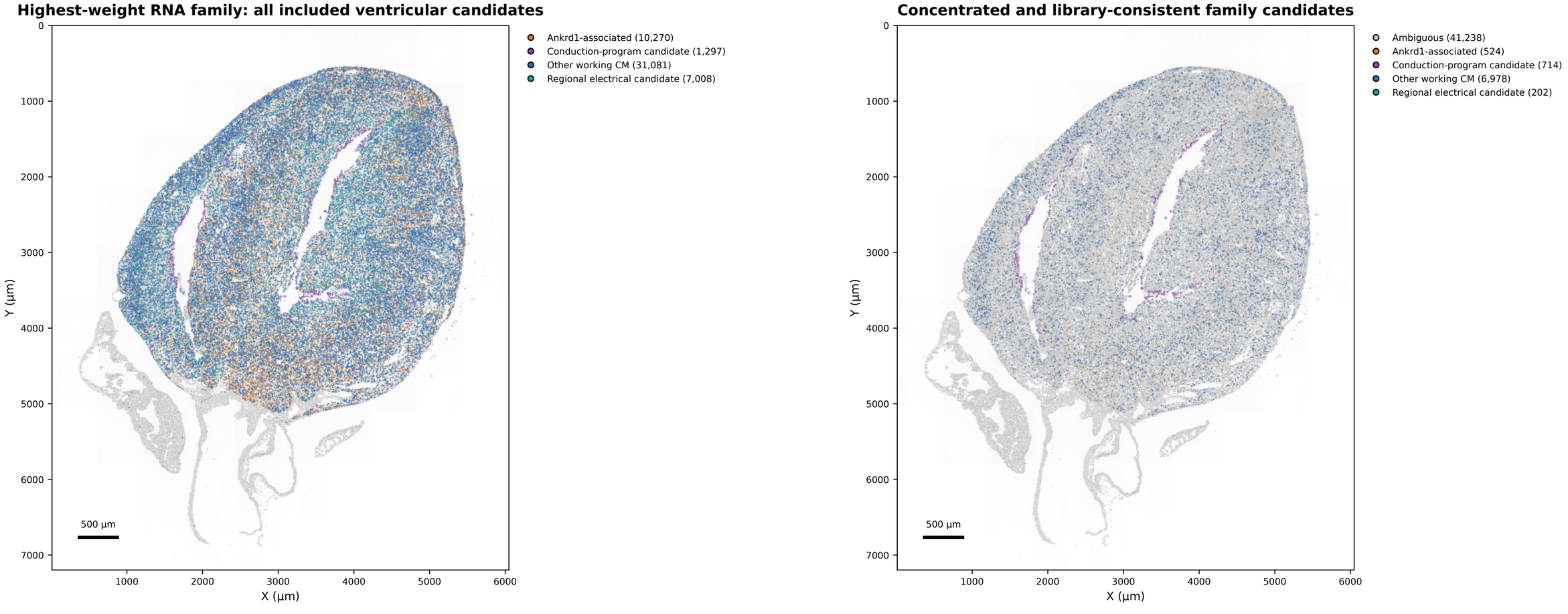


Concentrated and library-consistent family candidates



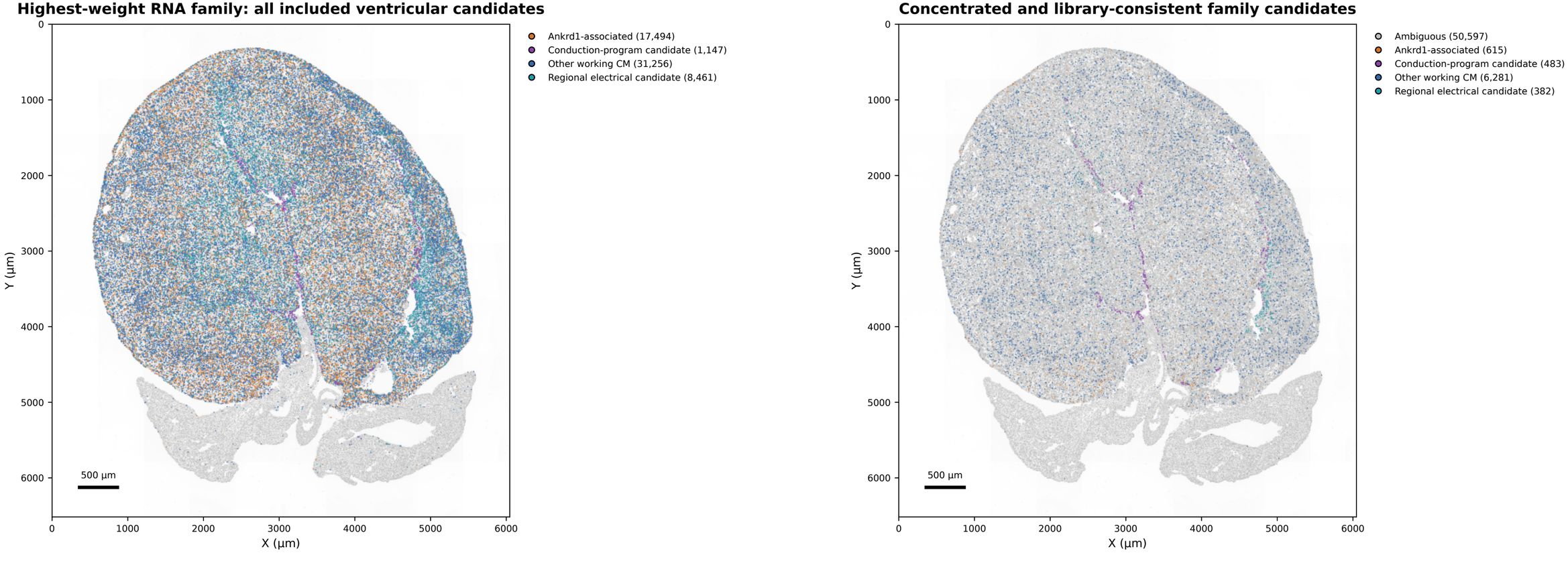
Original Xenium coordinates and counts are preserved. Left: strongest reference family; right: ≥0.70 neighbor weight and the same family from both RNA libraries.
Gray cells are ambiguous or outside the frozen cohort. Weight fractions are not calibrated probabilities. These flags differ from the previous CytoSPACE acceptance rules.
Conduction-program candidates are not verified Purkinje cells. Kcnd2/Irx5 are unassayed; neither is used to orient these sections.

P14: unsmoothed weighted projection of ventricular RNA programs



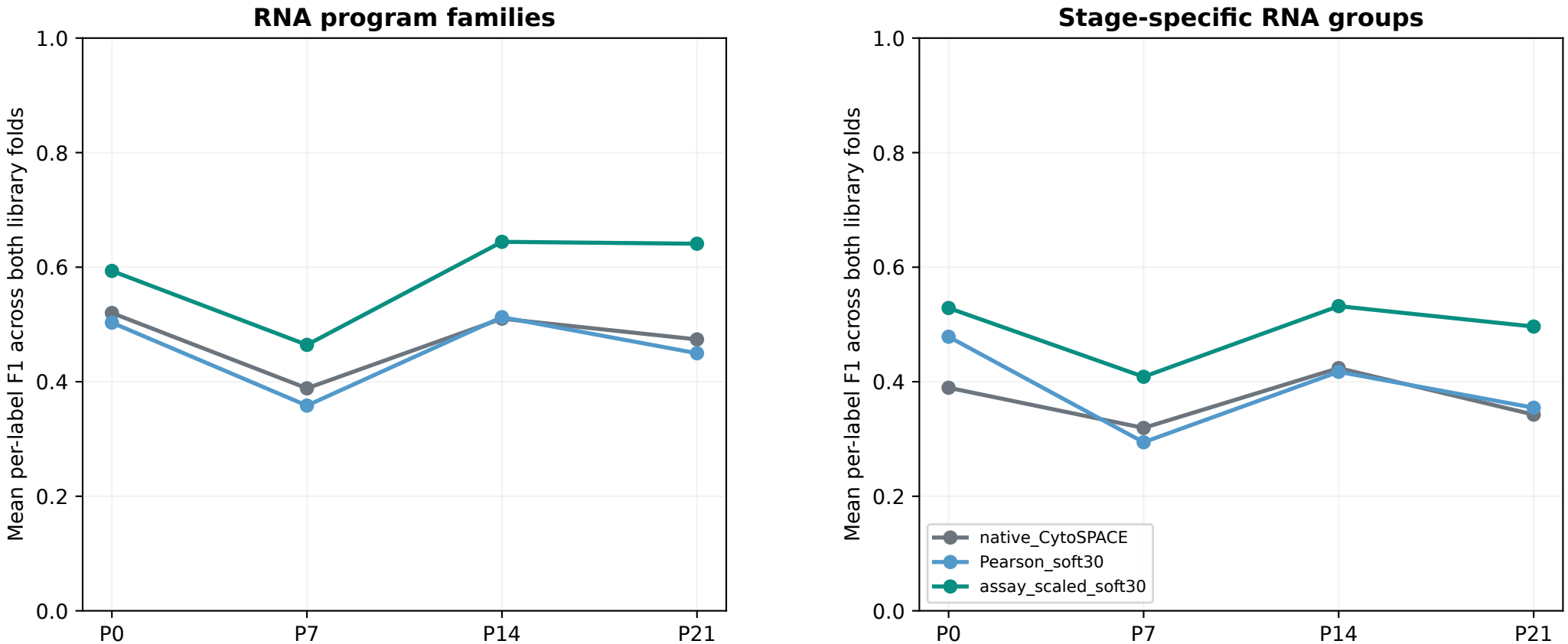
Original Xenium coordinates and counts are preserved. Left: strongest reference family; right: ≥0.70 neighbor weight and the same family from both RNA libraries.
Gray cells are ambiguous or outside the frozen cohort. Weight fractions are not calibrated probabilities. These flags differ from the previous CytoSPACE acceptance rules.
Conduction-program candidates are not verified Purkinje cells. Kcnd2/Irx5 are unassayed; neither is used to orient these sections.

P21: unsmoothed weighted projection of ventricular RNA programs



Original Xenium coordinates and counts are preserved. Left: strongest reference family; right: ≥0.70 neighbor weight and the same family from both RNA libraries.
Gray cells are ambiguous or outside the frozen cohort. Weight fractions are not calibrated probabilities. These flags differ from the previous CytoSPACE acceptance rules.
Conduction-program candidates are not verified Purkinje cells. Kcnd2/Irx5 are unassayed; neither is used to orient these sections.

Technical transfer between the two RNA libraries at each age



All381 shared genes; each query nucleus is excluded from its opposite-library reference. Existing RNA labels are not independent biological truth.
Native CytoSPACE values come from the preceding same-panel library benchmark; input ordering/partition realization can differ.